

When Do World-Model Signals Help VLAs?

Two Root Causes, Two One-Line Fixes

Anonymous submission (non-archival workshop draft v0.1)

Abstract—Conditioning vision-language-action (VLA) policies on world-model (WM) predictions promises task-level guidance for long-horizon manipulation, yet such signals rarely improve aggregate success, and several recent systems report that naive injection actively hurts. We present a diagnosis with per-task resolution. *Phenomenon*: subgoal conditioning redistributes per-task success — guidance-limited tasks gain, others lose — while every global mixing mechanism we tested leaves the aggregate on a plateau; a single task can drive an entire suite-level regression (76%→42%). *Root causes*: (i) *target redundancy* — across 1,693 episodes, ~95% of an absolute future target is explained by the current observation (residual-to-absolute discriminability 15×, 100% of episodes consistent); (ii) *WM-gradient interference* — prediction gradients flowing into the shared encoder degrade the policy, shown by intervention: detaching them recovers a regressed suite from 93.2 to 95.9 without hurting the primary suite. *Fixes, one line each*: residual targets (subgoal – current) lift LIBERO-10 to 95.75±0.76% ($n=8$; +1.55 over a multi-seed baseline, Welch $t\approx 2.8$); gradient isolation repairs the regression. We also report two of our own pre-registered hypotheses that *failed* — a retrieval-aliasing account of the single-task regression and a confusion-score predictability law — and argue that task-level units, multi-seed error bars, and pre-registered falsification lines should become defaults for this literature.

I. INTRODUCTION

Future-prediction signals are rapidly becoming a standard ingredient of VLA training: compressed latent future queries (DreamVLA[V:2507.04447]), future patch tokens in the policy’s own feature space (AHEAD[V:2606.02486]), generative future frames as auxiliary supervision (WorldVLA[V:2506.21539]), gated visuomotor future tokens (FutureVLA[V:2603.10712]), and retrieval-based progress signals (UR-VC[V:2607.12892]). Yet a striking pattern runs through this literature’s own ablations: *naive injection hurts*. DreamVLA reports that depth- or semantics-only future supervision falls below a no-prediction baseline (“competing gradients dilute the task-relevant features”); FutureVLA reports that adding future-motor tokens without gating *drops* success from 62.5% to 58.4%; WorldVLA finds that attending the model’s own past actions degrades grasping by 10–50 points unless masked. Third-party benchmarks echo the pattern at the task level: in WorldArena 2.0[V:primary source], the highest-fidelity predictor scores 100% on one manipulation task and 0% on another where a no-prediction baseline reaches 80%. The field has responded with local patches — attention masks, learnable gates, loss re-weighting — but without a diagnosis of *why* the signal harms.

This paper contributes the diagnosis, and a target design derived from it. Our claims:

- 1) **A task-level anatomy of the failure (§II)**: global mixing mechanisms (shared/dedicated queries, guidance scaling, phase-scheduled dropout) only redistribute per-task success; suite aggregates hide single-task collapses.
- 2) **Root cause I: target redundancy, measured at corpus scale (§III)** — ~95% of absolute future targets is current-state re-encoding (1,693 episodes, 100% consistent) — and its one-line fix, *residual targets*, absent from all six recent future-conditioned VLAs we surveyed.
- 3) **Root cause II: WM-gradient interference, established by intervention (§V)** — detaching WM prediction gradients from the shared encoder recovers a regressed suite (93.2→95.9) without hurting the primary one; this is a controlled confirmation of the “competing gradients” effect reported anecdotally by DreamVLA[V:].
- 4) **Negative results under pre-registration (§VI)**: a retrieval-aliasing mechanism and a confusion-score predictability law, both falsified by our own pre-registered tests — reported in full.

II. THE PRECISION–GUIDANCE TENSION

Setup. Our primary stack is a dual-channel latent-world-model VLA: a local channel predicts a fixed short-horizon future ($t+7$) in a frozen visual-semantic feature space, and a global channel predicts a *milestone* — a canonical, cross-episode-recurrent future state mined by retrieval over demonstrations. Both channels condition a flow-matching action expert. The second stack is a $\pi_{0.5}$ -style VLA[V:] that consumes the milestone signal as an extra conditioning token. Full training details in the appendix [TODO: appendix or cite tech report].

Observation. Table I summarizes the LIBERO-10 multi-seed picture. Naive milestone injection lifts the hardest guidance-limited task (t6, a two-object sequencing task) while degrading precision-limited tasks (t7, t8); dedicated query capacity (2Q) restores t8 but starves the guidance channel (t6 drops); guidance scaling (CFG) and diffusion-phase-scheduled dropout each shift the balance without breaking the plateau: every variant lands within $\sim 1\sigma$ of the multi-seed baseline aggregate (94.20±0.97). Inference-time knobs were falsified twice. A qualitatively similar trade appears on a second, $\pi_{0.5}$ -style stack (external-encoder hints severely harm its hardest task while own-encoder hints are benign), though with too few tasks to support a quantitative law — see §VI.

TABLE I
LIBERO-10, MULTI-SEED ($\pm$ POPULATION STD). EVERY GLOBAL MIXING LANDS ON A PLATEAU; PER-TASK STRUCTURE IS WHERE THE ACTION IS.

Variant	Agg.	t6 (guid.)	t8 (prec.)	t7 (prec.)
Baseline (local only, $n=4$)	94.20 $\pm$ 0.97	82.5	89.5 $\pm$ 12.4	100
+ milestone (naive)	94.6[V:]	90[V:]	84[V:]	96
+ dual channels	94.8	88	90	100
+ dedicated queries (2Q)	94.80 $\pm$ 0.71	85.0	88.5	98
+ phase-sched. drop ($n=8$)	95.22 $\pm$ 0.91	85.0	90.0	100
+ residual ($n=8$)	95.75$\pm$0.76	87.5	87.0	100

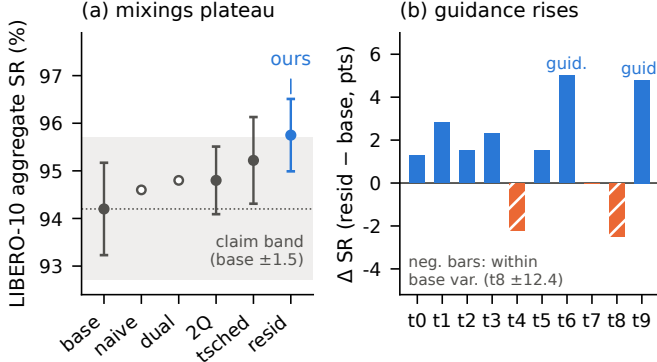


Fig. 1. (a) Aggregate plateau on LIBERO-10: every global mixing lands inside the baseline’s ± 1.5 -point claim band (shaded); only residual targets clear it (open markers: single-seed runs). (b) Per-task delta of residual vs. baseline: guidance tasks rise (+5.0/+4.8), precision tasks stay (t7 at 100; t8 within the baseline’s own ± 12.4 variance).

III. TWO ROOT CAUSES

A. Root cause I: target redundancy

Absolute subgoal targets are dominated by the present. Across the full corpus (1,693 episodes, four suites), $\sim 95\%$ of the target’s energy is explained by the current state — consistent in 100% of episodes — and the along-trajectory discriminability of the residual component $ms - h_t$ exceeds that of the absolute target by $\sim 15\times$ (per-episode examples: CV 1.1 vs. 0.12). Architecturally the redundancy is double: the conditioning stream already contains the current tokens h_t , so an absolute target re-encodes information that is already present — at best wasted capacity, at worst competing gradients, exactly the failure DreamVLA’s ablation describes[V:].

B. A single task drives a suite-level regression

The LIBERO-spatial suite provides a natural experiment. Under absolute milestone targets the suite regresses by -3.0 points — but per-task resolution shows the regression is *entirely one task* (Fig. ??): t5, “pick up the black bowl on the ramekin,” collapsing from 76% to 42% (18% in an independently retrained model), while the other nine tasks are unaffected (97–100% for every variant). Residual targets partially rescue t5 (52%).

We initially attributed this to cross-episode retrieval aliasing (t5 shares its object set with two near-duplicate tasks). A pre-registered confusion-score test *falsified* that account (§VI).

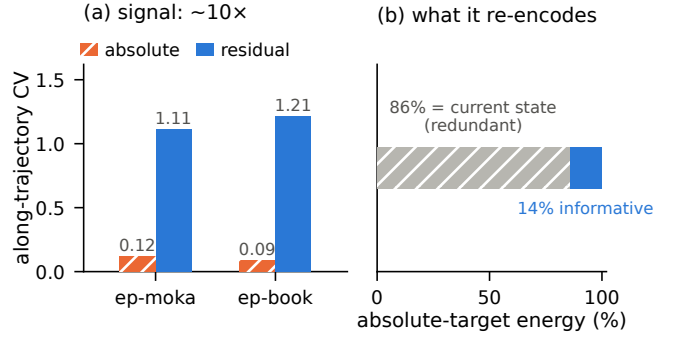


Fig. 2. Root cause I, measured. (a) Along-trajectory coefficient of variation of the injected target on two representative episodes: the residual carries $\sim 10\times$ the temporal signal of the absolute target. (b) Decomposition of the absolute target: 86% of its energy re-encodes the current state that the conditioning stream already contains.

The intervention that does repair the suite is *gradient isolation* (§V), implicating training-time interference rather than retrieval confusion.

IV. THE FIX: RESIDUAL TARGETS

The global channel’s ground truth becomes $h_{gt}^{ms} = ms - h_t$ instead of ms . The channel’s prediction (now a delta) conditions the action expert alongside the untouched current tokens h_t — a non-redundant decomposition: h_t carries “where I am,” the channel carries only “what must change.” The change is one line, parameter-free, and training/inference-consistent. Each element of the final design answers one diagnosed cause: dual parallel channels (tension: never overwrite the local precision channel), residual targets (redundancy), dedicated queries (capacity), and — suggested by our anatomy but left to future work — retrieval-confidence gating (aliasing).

V. ROOT CAUSE II: WM-GRADIENT INTERFERENCE, AND ITS FIX

The WM prediction heads are trained through the same encoder that serves the policy. An A/B intervention isolates the effect: detaching the WM-prediction gradients from the shared encoder (one line) recovers the regressed spatial suite from 93.2 to 95.9 — near the no-WM level of 96.5 — while leaving LIBERO-10 unharmed. The conditioning *signal* is preserved; only the gradient path changes. This turns the “competing gradients dilute task-relevant features” observation that DreamVLA[V:] reports anecdotally into a controlled, per-suite-verified mechanism, and reattributes our single-task regression (§III-B) to training-time interference. [TODO: exact per-task t5 number under detach; final table when frozen]

VI. WHAT WE FALSIFIED (PRE-REGISTERED)

Two of our own hypotheses died by their pre-registered criteria, and we report them as results. (1) *Retrieval-aliasing mechanism*: we predicted the collapsed task t5 would be an extreme outlier in a cross-episode confusion score; it is not, and the most-confused task triplet in another suite

shows zero harm. (2) *Predictability law*: we predicted per-task deltas would correlate with data-computable confusion/structure scores; the correlation does not survive (a difficulty–harm correlation on a second stack likewise loses significance once a single outlier task is removed). These negatives sharpen the paper’s scope: the defensible root causes are redundancy and gradient interference — both established by measurement or intervention, not narrative.

VII. RESULTS

Protocol. Multi-seed everywhere (population std reported; n per row); aggregate differences below 1.5 points are never claimed (per-seed binomial noise makes smaller aggregate deltas unattributable); pre-registered predictions with falsification lines were logged before each decisive evaluation [TODO: cite repo/appendix].

LIBERO-10. Residual targets give the only variant that clears the claim line: $95.75 \pm 0.76\%$ ($n=8$) versus 94.20 ± 0.97 ($n=4$), $\Delta = +1.55$, Welch $t \approx 2.8$. The shape matches the diagnosis: guidance tasks rise (t6 $82.5 \rightarrow 87.5$, $+5.0$; t9 $86.0 \rightarrow 90.8$, $+4.8$), precision tasks do not fall (t7 100 ± 0 ; t8 within its own large variance). Notably, an earlier pre-registered prediction for the spatial rescue (“ ≥ 95 ”) was *missed* (94.4): we report this as a partial confirmation — the mechanism is directionally right and quantitatively incomplete, which is precisely what motivates the two-component harm model of §III-B.

RoboTwin (blocks family, 6 tasks). With identical recipes and training steps, milestone conditioning turns a 46.4 ± 0.8 baseline into 49.3 ± 1.5 ($n=4$, $t=3.1$); the gain concentrates in the one multi-stage timing task (beat-block-hammer, $0.5\% \rightarrow 20.0\%$), while saturated tasks are flat and two zero-competence tasks (stacking) remain at 0% for both — subgoal signals need a competent base to have leverage. *Caveat (honest)*: the shared baseline is under-trained on this family relative to a single-task reference (90%); the A/B contrast is valid, absolute numbers are not leaderboard-grade [TODO: replace with retrained-to-convergence numbers when P1-E lands].

Scoping. On saturated suites (LIBERO-goal/object) all variants are flat [TODO: fill P1-F numbers]; on the aliased spatial task the fix is partial (above). Gains are selective — and, we argue, predictable: a task-level law fitting per-task deltas to data-computable subgoal structure and aliasing scores is our current pre-registered experiment [TODO: fill PR-2 verdict when available].

VIII. RELATED WORK

Future-signal injection spans four families: appended queries with structured attention masks (DreamVLA[V:]), future-token replacement with uncertainty-truncated adaptive horizons (AHEAD[V:] — the only adaptive-horizon design we are aware of), auxiliary-loss-only supervision (WorldVLA[V:], Spatial Forcing[V:2510.12276]), and gated cross-attention (FutureVLA[V:]); retrieval-based progress correction (UR-VC) forms a fifth, scalar, line. All predict *absolute* futures; task-level analysis stops at the suite granularity in all of them.

Their own ablations (competing gradients; naive-injection drops; action-history contamination) are, in our reading, three independent encounters with the redundancy/aliasing causes we isolate — patched respectively with masks, gates, and masking, i.e., treatments for a disease that had not been named.

IX. DISCUSSION AND OPEN PROBLEMS

(1) *The two fixes compose.* Residual targets act on what the channel predicts; gradient isolation acts on how it trains — their combination is our current experiment. (2) *Control-limited tasks do not respond to subgoals.* Our hardest precision task fails in grasp/place execution regardless of guidance — consistent with this workshop’s open problem of long-horizon force-control stability; the lever there is control-side (e.g., prefix-continuity replanning), which we find composes additively with signal-side fixes. (3) *Prediction quality is not downstream utility.* The WorldArena 2.0 reversal (best-fidelity predictor: 100% on one task, 0% where a no-prediction baseline gets 80%) mirrors our t5 anatomy; both argue that the field’s bottleneck is *what* to predict, not how well. (4) *Claim discipline.* Multi-seed error bars, task-level units, and pre-registered falsification lines changed our own conclusions twice (a single-seed baseline mirage; a partial-rescue verdict); we suggest they become community defaults for this literature.

REFERENCES

- [1] V. TBD, “Ur-vc (verify full title),” *arXiv preprint arXiv:2607.12892*, 2026, vERIFY all fields before submission.
- [2] —, “Robodojo: A unified sim-and-real benchmark for comprehensive evaluation of generalist robot manipulation policies,” *arXiv preprint arXiv:2607.04434*, 2026, vERIFY all fields before submission.